

Kartik Madan

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London, United Kingdom

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EDUCATION

Guru Gobind Singh Indraprastha University

Bachelor of Engineering in Electronics and Communication; GPA: 3.28 (8.2/10.0)

New Delhi, India
2016 - 2020

EXPERIENCE

BP p.l.c.

Sep 2024 - Present

Conversational Analytics Platform for Trading Operations

London, United Kingdom

Software Engineer

- Delivered a conversational analytics platform end to end: React/TypeScript frontend, FastAPI backend, authentication and session management, CI/CD pipelines and AWS CDK infrastructure-as-code, cutting query turnaround from hours to seconds.
- Designed the backend services and REST APIs that let trading operators query BP's trading warehouse, orchestrating a four-stage request pipeline over enterprise datasets.
- Built an agent-orchestration layer over large language models (AWS Bedrock) that turns operators' natural-language questions into accurate, governed SQL, grounded in business rules, warehouse metadata and domain terminology.
- Engineered a query safety and validation layer (read-only enforcement, row limits, injection protection) to guarantee the correctness and security of every response.
- Built automated unit, integration and end-to-end tests across frontend and backend behind a CI coverage gate, along with an AI evaluation harness scoring text-to-SQL accuracy against known cases.

Contract Value Leakage

London, United Kingdom

Software Engineer

- Built document-processing pipelines and backend services reconciling ~15K supplier invoices/month against agreed price books.
- Developed extraction and validation services spanning 200+ supplier templates and 50+ invoice formats at 95%+ accuracy.
- Implemented an invoice-to-contract matching and anomaly-detection engine, reducing undetected contract leakage by ~40%.

Wild Bean Cafe

London, United Kingdom

Full Stack Engineer / Project Lead

- Led development of a cloud-native pricing, sales and inventory platform for self-service Micro Market units across 100+ BP forecourt retailers in the UK and Poland.
- Partnered with POS, SAP and platform teams to build the enterprise integration and analytics workflows connecting in-store systems to cloud data platforms.
- Added natural-language analytics letting retail managers query operational data directly.
- Contributed to a 148% improvement in dealer profitability through automation and reporting.

Store of Today

Gurgaon, India

Full Stack Engineer

- Built event-driven services for real-time hazard detection using computer vision across retail locations.
- Developed dynamic pricing over Electronic Shelf Labels (ESLs).
- Automated AWS infrastructure and deployment pipelines with Terraform.

Infosys

Nov 2020 - Sep 2023

Equinox Studio

Gurgaon, India

Full Stack Engineer

- Built a low-code/no-code digital commerce platform used by global clients (Kraft Heinz, Shutterfly, Herbalife, Nu Skin), contributing 22% of the team's revenue.

PROJECTS & OPEN SOURCE

- Energy Trading:** Market-participation platform for India's power-trading ecosystem, with ML-driven forecasting.
- Price Forecasting Engine:** RNN-based deep-learning app that forecasts market price movements and streams predictions into live trading platforms.
- Microsoft Agent Framework:** Open-source contributor to Microsoft's framework for building and orchestrating agent workflows.
- ContractIQ:** System to create and analyze legal contracts, built on LangChain.

SKILLS

- Languages:** Python, Java, TypeScript, JavaScript, C++, C#, SQL
- Frontend:** React, Next.js, Vue, HTML/CSS
- Backend:** FastAPI, Node.js, REST APIs
- Cloud & DevOps:** AWS, GCP, Terraform, Docker, CI/CD (Azure DevOps)
- Data:** PostgreSQL, DynamoDB, OpenSearch, BigQuery, SQL/NoSQL
- AI/ML:** LLMs, RAG, Agentic AI, TensorFlow

ACHIEVEMENTS

- Codeforces:** World Rank 625, National Rank 37 (of 7,585) at best finish.
- LeetCode:** Knight, 1,854 contest rating.
- Hackathons:** Winner, Smart India Hackathon 2019; Winner, Barco Geekathon 2018.
- Infosys RISE Awards:** Business Ninja (Q3 FY24), Tech Wizard (Q4 FY23).

PUBLICATIONS

- Patent (US 2026/0141595, published):** System and method for identifying and replacing proprietary images with synthetically generated images.
- Springer, ICICC 2020:** Development of a real-time pollution monitoring system for green auditing.